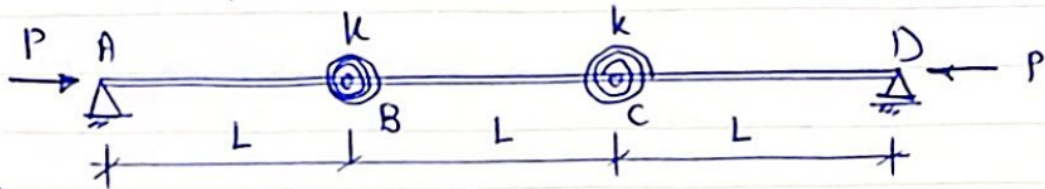
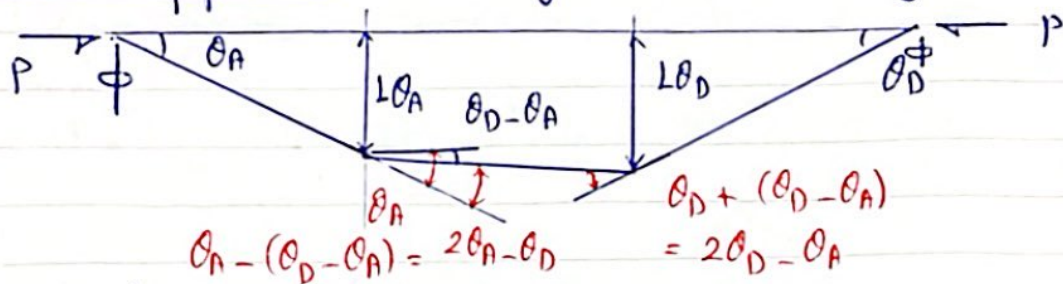


Example (2):- Compute P_{cr} using bifurcation and Energy approach.



Solution:-

Bifurcation approach 2 Degree of freedom system.

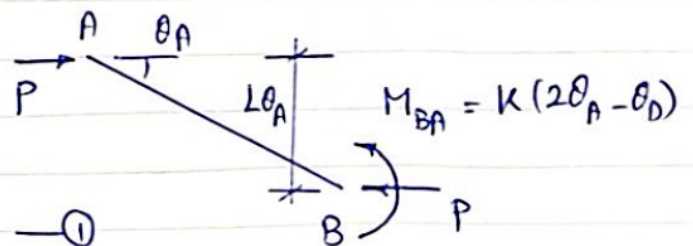


Free body diagram of AB

$$\sum M_A = 0.0$$

$$M_{BA} - PL\theta_A = 0.0$$

$$\therefore K(2\theta_A - \theta_D) - PL\theta_A = 0.0 \quad \text{--- (1)}$$

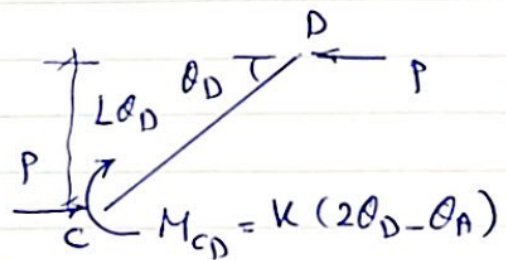


Free body diagram of CD

$$\sum M_D = 0.0$$

$$M_{CD} - PL\theta_D = 0.0$$

$$\therefore K(2\theta_D - \theta_A) - PL\theta_D = 0.0 \quad \text{--- (2)}$$



Rearrange ① & ② in a matrix form.

$$\begin{bmatrix} 2K - PL & -K \\ -K & 2K - PL \end{bmatrix} \begin{Bmatrix} \theta_A \\ \theta_D \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

for non-trivial solution $|K| = 0.0$

$$\therefore (2k - PL)^2 - k^2 = 0.0 \quad (2)$$

$$\therefore 4k^2 - 4kPL + P^2L^2 - k^2 = 0.0$$

$$\therefore 3k^2 - 4kPL + P^2L^2 = 0.0$$

$$\therefore (k - PL)(3k - PL) = 0.0$$

$$\therefore P = \frac{k}{L}, \frac{3k}{L} = \text{eigen values}$$

$\therefore P_{cr} = \text{smaller eigen value} = (k/L)$.

To find eigen vectors substitute $P = k/L, 3k/L$ in equilibrium equations.

$$\text{let } P = k/L \quad \therefore (2k - k)\theta_A - k\theta_D = 0.0 \quad \therefore [\theta_A = \theta_D]$$



$$\text{let } P = 3k/L \quad \therefore (2k - 3k)\theta_A - k\theta_D = 0.0 \quad \therefore [\theta_A = -\theta_D]$$

Energy approach:-

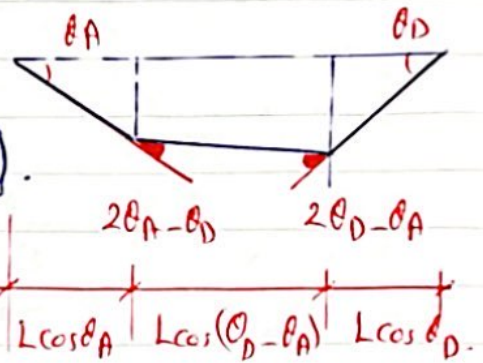
$$\Pi = U + W$$

$$U = \frac{1}{2} k (2\theta_A - \theta_D)^2 + \frac{1}{2} k (2\theta_D - \theta_A)^2$$

$$W = -PL [3 - \cos\theta_A - \cos(\theta_D - \theta_A) - \cos\theta_D].$$



$$\therefore \Pi = \frac{1}{2} k (2\theta_A - \theta_D)^2 + \frac{1}{2} k (2\theta_D - \theta_A)^2 - PL (3 - \cos\theta_A - \cos(\theta_D - \theta_A) - \cos\theta_D)$$



Condition of equilibrium:- $\delta\Pi = 0.0 \quad \therefore \frac{\delta\Pi}{\delta\theta_A} = 0.0 \quad \therefore \frac{\delta\Pi}{\delta\theta_D} = 0.0$

Condition of Stability:- $\delta^2\Pi > 0.0 \quad \therefore \rightarrow \frac{\delta^2\Pi}{\delta\theta_A^2} > 0.0$

$$\rightarrow \frac{\delta^2\Pi}{\delta\theta_D^2} > 0.0$$

$$\rightarrow \left(\frac{\delta^2\Pi}{\delta\theta_A^2} \cdot \frac{\delta^2\Pi}{\delta\theta_D^2} \right) > \left(\frac{\delta^2\Pi}{\delta\theta_A \cdot \delta\theta_D} \right)^2$$

Proof:-

(3)

$$\begin{aligned} \Pi(A+\Delta A, B+\Delta B) &= \Pi(A, B) + \frac{1}{1!} \left[\frac{\partial \Pi}{\partial A} \cdot \Delta A + \frac{\partial \Pi}{\partial B} \cdot \Delta B \right] \\ &+ \frac{1}{2!} \left[\frac{\partial^2 \Pi}{\partial A^2} \Delta A^2 + \frac{2 \partial^2 \Pi}{\partial A \partial B} \Delta A \Delta B + \frac{\partial^2 \Pi}{\partial B^2} \Delta B^2 \right] \\ &+ \frac{1}{3!} \left[\frac{\partial^3 \Pi}{\partial A^3} \Delta A^3 + 2 \frac{\partial^3 \Pi}{\partial A^2 \partial B} \Delta A^2 \Delta B + 2 \frac{\partial^3 \Pi}{\partial B^2 \partial A} \Delta B^2 \Delta A \right. \\ &\left. + \frac{\partial^3 \Pi}{\partial B^3} \Delta B^3 \right] + \dots \end{aligned}$$

$$\Pi(A+\Delta A, B+\Delta B) = \Pi(A, B) + \frac{1}{1!} \delta \Pi + \frac{1}{2!} \delta^2 \Pi + \frac{1}{3!} \delta^3 \Pi + \dots$$

First variation, $\delta \Pi = \frac{\partial \Pi}{\partial A} \cdot \Delta A + \frac{\partial \Pi}{\partial B} \cdot \Delta B = 0.0$

Since ΔA & ΔB are independent

$\therefore \frac{\partial \Pi}{\partial A} = 0.0$, $\frac{\partial \Pi}{\partial B} = 0.0$ equilibrium equations

Second Variation, $\delta^2 \Pi = \frac{\partial^2 \Pi}{\partial A^2} \Delta A^2 + 2 \frac{\partial^2 \Pi}{\partial A \partial B} \Delta A \Delta B + \frac{\partial^2 \Pi}{\partial B^2} \Delta B^2$

$$= (\Delta A \quad \Delta B) \begin{bmatrix} \frac{\partial^2 \Pi}{\partial A^2} & \frac{\partial^2 \Pi}{\partial A \partial B} \\ \frac{\partial^2 \Pi}{\partial A \partial B} & \frac{\partial^2 \Pi}{\partial B^2} \end{bmatrix} \begin{Bmatrix} \Delta A \\ \Delta B \end{Bmatrix}$$

for $|\det| > 0.0$

$$\therefore \frac{\partial^2 \Pi}{\partial A^2} > 0.0 \quad \text{--- (1)} \quad \frac{\partial^2 \Pi}{\partial B^2} > 0.0 \quad \text{--- (2)}$$

$$\therefore \left[\frac{\partial^2 \Pi}{\partial A^2} \cdot \frac{\partial^2 \Pi}{\partial B^2} \right] > \left(\frac{\partial^2 \Pi}{\partial A \partial B} \right)^2 \quad \text{--- (3)}$$

These are the three conditions required to be satisfied to prove that equilibrium is stable.

$$\therefore \frac{\partial \Pi}{\partial \theta_A} = 2k(2\theta_A - \theta_D) - k(2\theta_D - \theta_A) - PL(\sin \theta_A + \sin(\theta_D - \theta_A)) = 0.0$$

$$\frac{\partial \Pi}{\partial \theta_D} = -k(2\theta_A - \theta_D) + 2k(2\theta_D - \theta_A) - PL(\sin \theta_D - \sin(\theta_D - \theta_A)) = 0.0$$

let $\sin \theta \approx \theta$ & rewrite in a matrix form.

$$\therefore \begin{bmatrix} (5k - 2PL) & (-4k + PL) \\ (-4k + PL) & (5k - 2PL) \end{bmatrix} \begin{bmatrix} \theta_A \\ \theta_D \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

for a non-trivial solution $|K| = 0.0$

$$\therefore (5k - 2PL)^2 - (-4k + PL)^2 = 0.0 \quad \therefore P = \frac{k}{L}, \frac{3k}{L}$$

$$\therefore P_{cr} = \text{smaller eigenvalue} = \left(\frac{k}{L}\right) \quad \text{eigen value}$$

To study type of equilibrium:

$$\textcircled{1} \frac{\partial^2 \Pi}{\partial \theta_A^2} = 5k - 2PL \quad \therefore \text{for stable equilib.} \quad P \leq \left(\frac{2.5k}{L}\right)$$

$$\textcircled{2} \frac{\partial^2 \Pi}{\partial \theta_D^2} = 5k - 2PL \quad \therefore \text{for stable equilib.} \quad \frac{\partial^2 \Pi}{\partial \theta_D^2} > 0.0$$

$$\therefore P \leq \left(\frac{2.5k}{L}\right).$$

$$\textcircled{3} \left(\frac{\partial^2 \Pi}{\partial \theta_A^2} \cdot \frac{\partial^2 \Pi}{\partial \theta_D^2} \right) > \left(\frac{\partial^2 \Pi}{\partial \theta_A \partial \theta_D} \right)^2$$

$$\therefore (5k - 2PL)^2 > (-4k + PL)^2 \quad \text{(governs)}$$

$$\therefore (k - PL)(3k - PL) > 0.0 \quad \therefore P \leq k/L, \quad P \leq \frac{3k}{L}$$

$\therefore$ Condition of Stable equilibrium $(P \leq k/L)$

Large deflection Analysis:-

(5)

- Using small deflection analysis = P_{cr} & buckled Shape. check Eq. at fundamental
- Using large deflection analysis = - P_{cr} & buckled Shape.
 - post-buckling path.
 - check equilibrium at
 - fundamental path.
 - critical load.
 - post buckling path.

Example:-

find P_{cr} , buckled Shape
postbuckling path
check equilibrium at fundamental path,
critical load and post-buckling path.

Solution:-

$$\Pi = U + W$$

$$U = \frac{1}{2} k \theta^2$$

$$W = -PL(1 - \cos \theta)$$

$$\therefore \Pi = \frac{1}{2} k \theta^2 - PL(1 - \cos \theta)$$

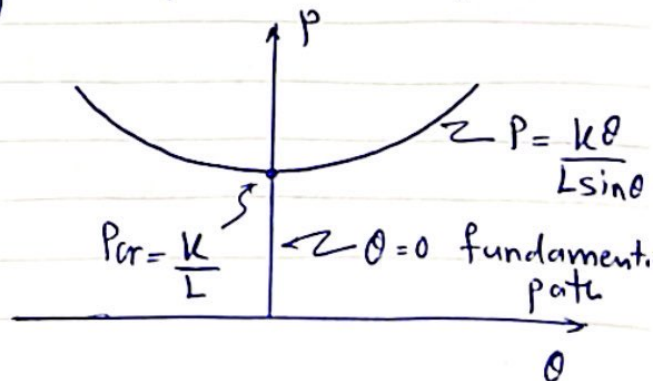
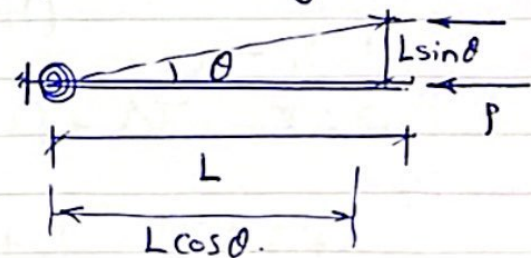
For equilibrium:- $\frac{d\Pi}{d\theta} = 0 \quad \therefore k\theta - PL \sin \theta = 0.0$

$\therefore \theta = 0.0 \leftarrow$ fundamental path.

$\therefore P = \frac{k\theta}{L \sin \theta} \leftarrow$ post-buckling path

To study stability, we need higher derivatives of Π

$$\frac{d^2\Pi}{d\theta^2} = (k - PL \cos \theta).$$



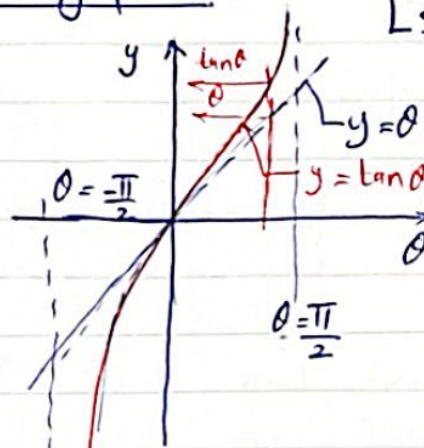
- check Stability at fundamental path, $\theta = 0$ (6)
 $\therefore \frac{d^2\pi}{d\theta^2} = K - PL$ $\therefore$ if $P < K/L$ $\frac{d^2\pi}{d\theta^2} = (+)$ Stable

if $P > K/L$ $\frac{d^2\pi}{d\theta^2} = (-)$ Unstable

- check Stability at post-buckling path, $P = \frac{K\theta}{L \sin \theta}$

$$\therefore \frac{d^2\pi}{d\theta^2} = K \left[1 - \frac{\theta}{\tan \theta} \right]$$

= (+ve) Stable



$\tan \theta > \theta$

- check at critical load,
 $(P = K/L \neq \theta = 0)$

$$\frac{d^2\pi}{d\theta^2} = K - K = 0.0 \text{ (not defined)}$$

from Taylor series $\pi(\theta + \Delta\theta) = \pi(\theta) + \Delta\pi$ $\Delta\pi$ is (fur) $\rightarrow$ stable
 $(-ve) \rightarrow$ Unstable

$$\therefore \Delta\pi = \frac{1}{1!} \frac{d\pi}{d\theta} \Delta\theta + \frac{1}{2!} \frac{d^2\pi}{d\theta^2} \Delta\theta^2 + \frac{1}{3!} \frac{d^3\pi}{d\theta^3} \Delta\theta^3 + \dots$$

always $\uparrow$
 zero from equilibrium.

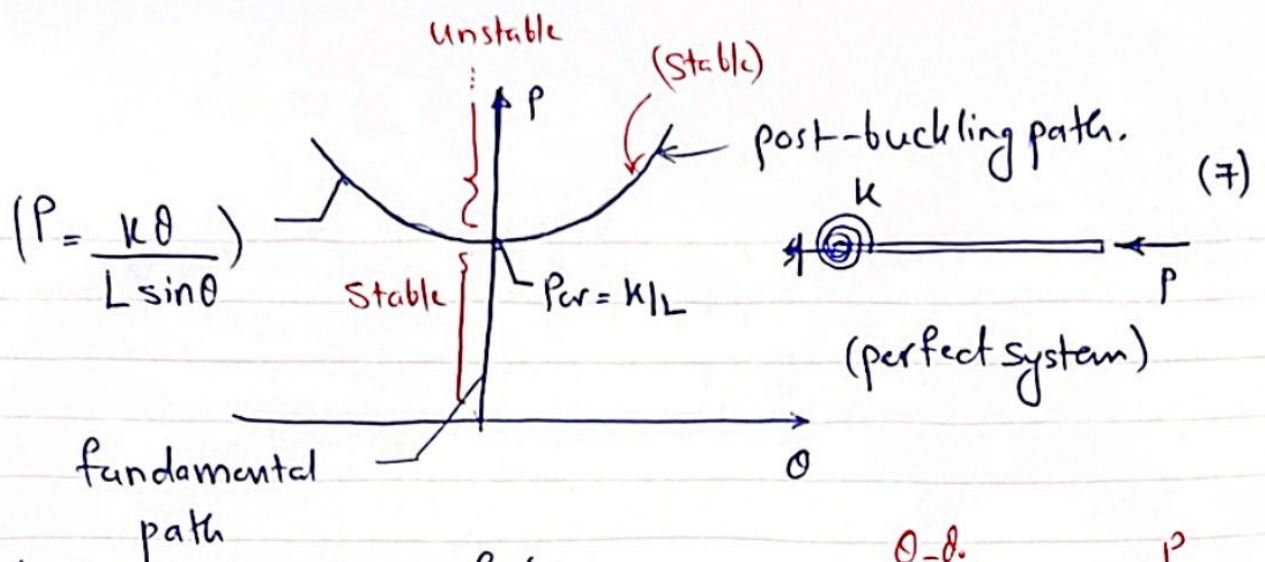
if $\frac{d^2\pi}{d\theta^2} = 0$, search for following terms in $\Delta\pi$.

$$\therefore \frac{d^3\pi}{d\theta^3} = PL \sin \theta \quad [\theta = 0, P = K/L] \quad \therefore \frac{d^3\pi}{d\theta^3} = 0.0$$

$$\therefore \frac{d^4\pi}{d\theta^4} = PL \cos \theta \quad \text{let } \theta = 0, P = K/L \quad \therefore \frac{d^4\pi}{d\theta^4} = (K) \text{ (+ve)}$$

$\therefore$ Stable at critical point.

$\therefore$ Stable.

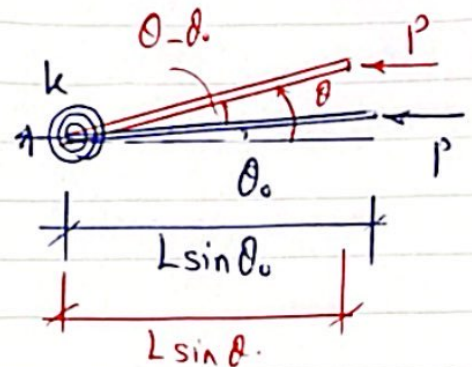


Effect of geometric imperfection:-

$$\Pi = U + W$$

$$U = \frac{1}{2} k (\theta - \theta_0)^2$$

$$W = -PL (\cos \theta_0 - \cos \theta)$$



$$\frac{d\Pi}{d\theta} = k(\theta - \theta_0) - PL \sin \theta = 0$$

only one solution is available

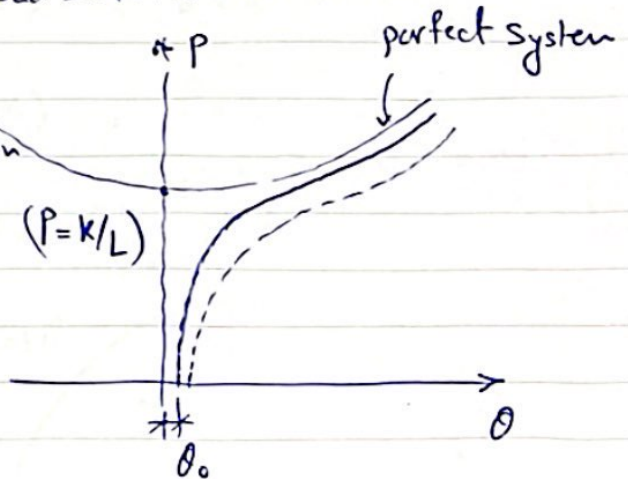
$$\therefore P = \frac{k(\theta - \theta_0)}{L \sin \theta} \quad \leftarrow \text{load-deflection curve}$$

check Stability along load-deflection Curve:-

$$\frac{d^2\Pi}{d\theta^2} = k - PL \cos \theta$$

$$\text{let } P = \frac{k(\theta - \theta_0)}{L \sin \theta}$$

$$\therefore \frac{d^2\Pi}{d\theta^2} = k \left[1 - \frac{\theta - \theta_0}{\tan \theta} \right]$$



$$\therefore \frac{\theta - \theta_0}{\tan \theta} < 1.0 \quad \therefore \frac{d^2\Pi}{d\theta^2} = (+ve) \text{ Stable equilibrium}$$

$\therefore$ This system is not sensitive to geometric imperfection

Example (2):- Find P_{cr} and buckled shape

(8)

- post-buckling path.
- check stability at fundamental path, critical load and post-buckling path
- Investigate effect of geometric imperfection.

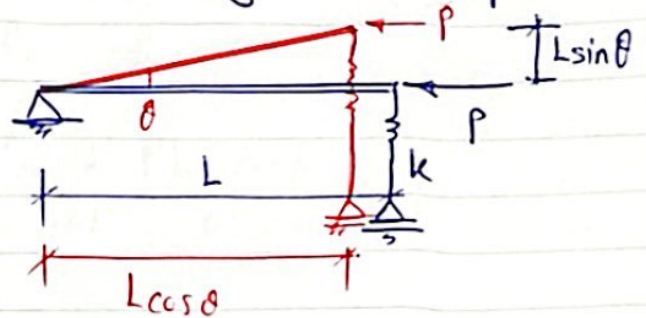
Solution:-

$$\Pi = U + W$$

$$U = \frac{1}{2} k (L \sin \theta)^2$$

$$W = -PL(1 - \cos \theta)$$

$$\therefore \Pi = \frac{1}{2} k L^2 \sin^2 \theta - PL(1 - \cos \theta)$$



Equilibrium: $\frac{d\Pi}{d\theta} = k L^2 \sin \theta \cdot \cos \theta - PL \sin \theta = 0.0$

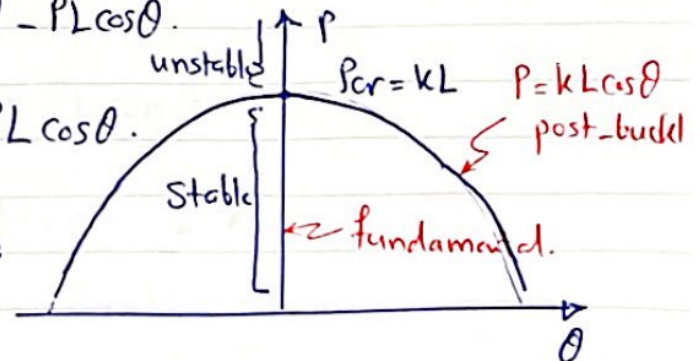
$$\therefore \frac{d\Pi}{d\theta} = (k L^2 \cos \theta - PL) \sin \theta = 0.0$$

$$\therefore \theta = 0.0 \text{ (fundamental path)}, \quad P = k L \cos \theta \text{ (post-buckling path)}$$

Stability:-

$$\frac{d^2\Pi}{d\theta^2} = k L^2 \cos^2 \theta - k L^2 \sin^2 \theta - PL \cos \theta$$

$$= k L^2 (\cos^2 \theta - \sin^2 \theta) - PL \cos \theta$$



check Stability along fundamental path:- $\theta = 0.0$

$$\frac{d^2\Pi}{d\theta^2} = k L - P \quad \therefore \text{if } P < k L \text{ Stable}$$

$$\text{if } P > k L \text{ Unstable}$$

check Stability along post-buckling path:- $P = k L \cos \theta$

$$\frac{d^2\Pi}{d\theta^2} = k L^2 (\cos^2 \theta - \sin^2 \theta) - k L^2 \cos \theta^2 = -k L^2 \sin^2 \theta$$

always (-ve) $\rightarrow$ Unstable

check Stability at critical point i.e.
 $\theta = 0$, $P = KL$

(9)

$$\therefore \frac{d^2\pi}{d\theta^2} = 0.0$$

$$\begin{aligned} \therefore \frac{d^3\pi}{d\theta^3} &= -2KL^2 \cos\theta \sin\theta - 2KL^2 \sin\theta \cos\theta + PL \sin\theta \\ &= -4KL^2 \cos\theta \sin\theta + PL \sin\theta \\ &= 0.0 \end{aligned}$$

$$\begin{aligned} \therefore \frac{d^4\pi}{d\theta^4} &= 4KL^2 \sin^2\theta - 4KL^2 \cos^2\theta + PL \cos\theta \\ &= -4KL^2 + KL^2 = -3KL^2 \text{ always (ve) Unstable.} \end{aligned}$$

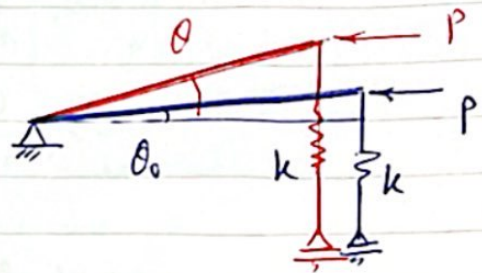
Effect of geometric imperfection:-

$$\pi = U + W$$

$$U = \frac{1}{2} k (L \sin\theta - L \sin\theta_0)^2$$

$$W = -PL (\cos\theta_0 - \cos\theta)$$

$$\therefore \pi = \frac{1}{2} KL^2 (\sin\theta - \sin\theta_0)^2 - PL (\cos\theta_0 - \cos\theta)$$



$$\frac{d\pi}{d\theta} = KL^2 (\sin\theta - \sin\theta_0) \cos\theta - PL \sin\theta = 0.0$$

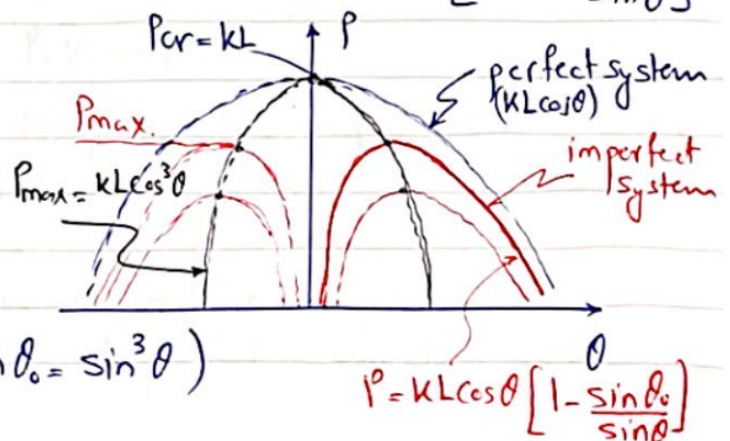
$$\therefore P = \frac{KL (\sin\theta - \sin\theta_0) \cos\theta}{\sin\theta} = KL \cos\theta \left[1 - \frac{\sin\theta_0}{\sin\theta} \right]$$

to find the maximum load

$$\frac{dP}{d\theta} = 0.0$$

$$= KL \left(-\sin\theta + \frac{\sin\theta_0}{\sin^2\theta} \right) = 0.0$$

$$\therefore P_{\max} \text{ takes place at } (\sin\theta_0 = \sin^3\theta)$$



$$\therefore P_{\max} = KL \cos \theta \left[1 - \frac{\sin^3 \theta}{\sin \theta} \right] = KL \cos \theta (1 - \sin^2 \theta) = KL \cos^3 \theta.$$

Check Stability

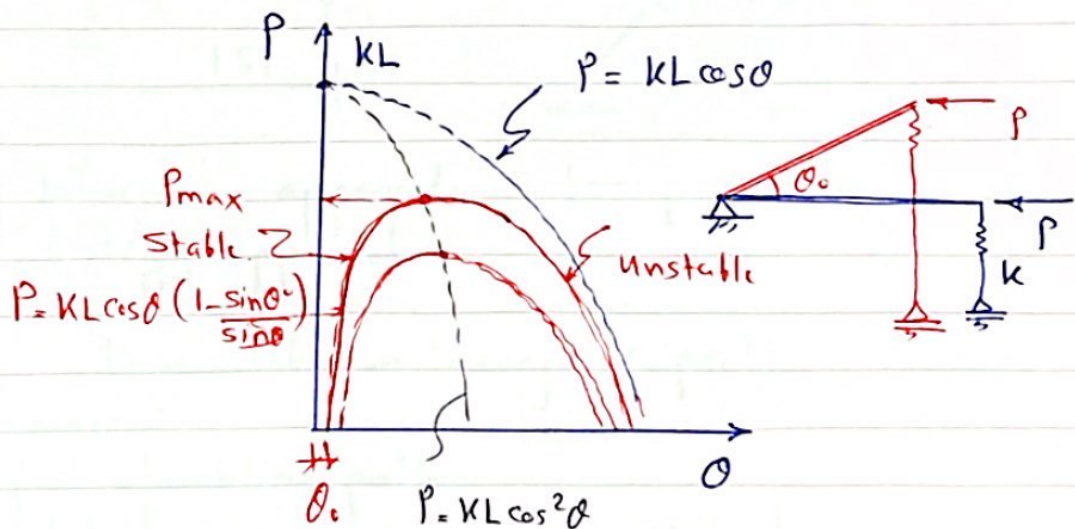
$$\frac{d^2 \Pi}{d\theta^2} = KL^2 (\cos^2 \theta - \sin^4 \theta + \sin \theta_0 \sin \theta) - PL \cos \theta.$$

$$\text{Let } P = KL \cos \theta \left(1 - \frac{\sin \theta_0}{\sin \theta} \right).$$

$$\therefore \frac{d^2 \Pi}{d\theta^2} = KL^2 \left(\frac{\sin \theta_0 - \sin^3 \theta}{\sin \theta} \right).$$

$$\text{if } \sin \theta_0 > \sin^3 \theta, \quad \frac{d^2 \Pi}{d\theta^2} > 0.0 \quad P < P_{\max} \quad \text{Stable}$$

$$\text{if } \sin \theta_0 < \sin^3 \theta, \quad \frac{d^2 \Pi}{d\theta^2} < 0.0 \quad P > P_{\max} \quad \text{Unstable}$$



- System is sensitive to geometric imperfection.
 $P_{cr}(\text{perfect}) \gg \text{imperfect}.$

- load - deflection Curve asymptotic to perfect system.

- if - post buckling path of perfect is unstable $\rightarrow$ sensitive to geometric imperfection.